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SCAPTURE: a deep learning-embedded pipeline that captures polyadenylation information from 3′ tag-based RNA-seq of single cells

Guo-Wei Li[1†], Fang Nan[1†], Guo-Hua Yuan[1], Chu-Xiao Liu[2], Xindong Liu[3], Ling-Ling Chen[2,4,5], Bin Tian[6] and Li Yang[1,4*]

* Correspondence: liyang@picb.ac.cn
[†]Guo-Wei Li and Fang Nan contributed equally to this work.
[1]CAS Key Laboratory of Computational Biology, Shanghai Institute of Nutrition and Health, University of Chinese Academy of Sciences, Chinese Academy of Sciences, Shanghai 200031, China
[4]School of Life Science and Technology, ShanghaiTech University, Shanghai 201210, China
Full list of author information is available at the end of the article

Abstract

Single-cell RNA-seq (scRNA-seq) profiles gene expression with high resolution. Here, we develop a stepwise computational method-called SCAPTURE to identify, evaluate, and quantify cleavage and polyadenylation sites (PASs) from 3′ tag-based scRNA-seq. SCAPTURE detects PASs de novo in single cells with high sensitivity and accuracy, enabling detection of previously unannotated PASs. Quantified alternative PAS transcripts refine cell identity analysis beyond gene expression, enriching information extracted from scRNA-seq data. Using SCAPTURE, we show changes of PAS usage in PBMCs from infected versus healthy individuals at single-cell resolution.

Keywords: scRNA-seq, PAS, APA, Deep learning, Peak calling, Transcript quantification

Introduction

The advent of single-cell RNA-seq (scRNA-seq) has enabled gene expression analysis with an unprecedented resolution [1, 2]. Based mainly on differential gene expression (DGE) [3], scRNA-seq reveals heterogeneity within a bulk of cells [4], complex tissues [5–8], or even the whole animal [9], resulting in the identification of distinct cell identities and lineage trajectories, especially in developing or differentiating systems [10, 11]. By taking advantage of machine-based cell isolation, hundreds of thousands of cells can now be individually processed for RNA enrichment and deep sequencing analysis [12, 13]. Recently, scRNA-seq technologies that use oligo(dT) priming for cDNA generation and subsequent short-read sequencing from their 3′-ends (herein called 3′ tag-based scRNA-seq) [14], such as inDrops [15], Drop-seq [12], Seq-Well [16], and 10x Chromium [13], have been broadly adopted. In contrast to full-length scRNA-seq (such as Smart-seq2) and canonical bulk cell RNA-seq (such as Illumina TruSeq), these 3′ tag-based scRNA-seq data characteristically have an enrichment of reads at the 3′

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ends of genes. For example, a comparison of transcriptome profiling in human PBMCs with TruSeq, Smart-seq2, and 10x Chromium [17] showed that a bias of mapped reads to annotated cleavage and polyadenylation sites (PASs) in the 10x Chromium data for both protein-coding (such as *GAPDH*, Fig. 1a, b) and noncoding (such as *NORAD* and *GAS5*, Additional file 1: Fig. S1A) genes. By contrast, TruSeq and Smart-seq2 data profile gene expression with reads covering the whole gene body (Fig. 1a, b; Additional file 1: Fig. S1B). As such, 3′ tag-based scRNA-seq datasets can be mined for PAS identification and for expression of transcripts using specific PASs.

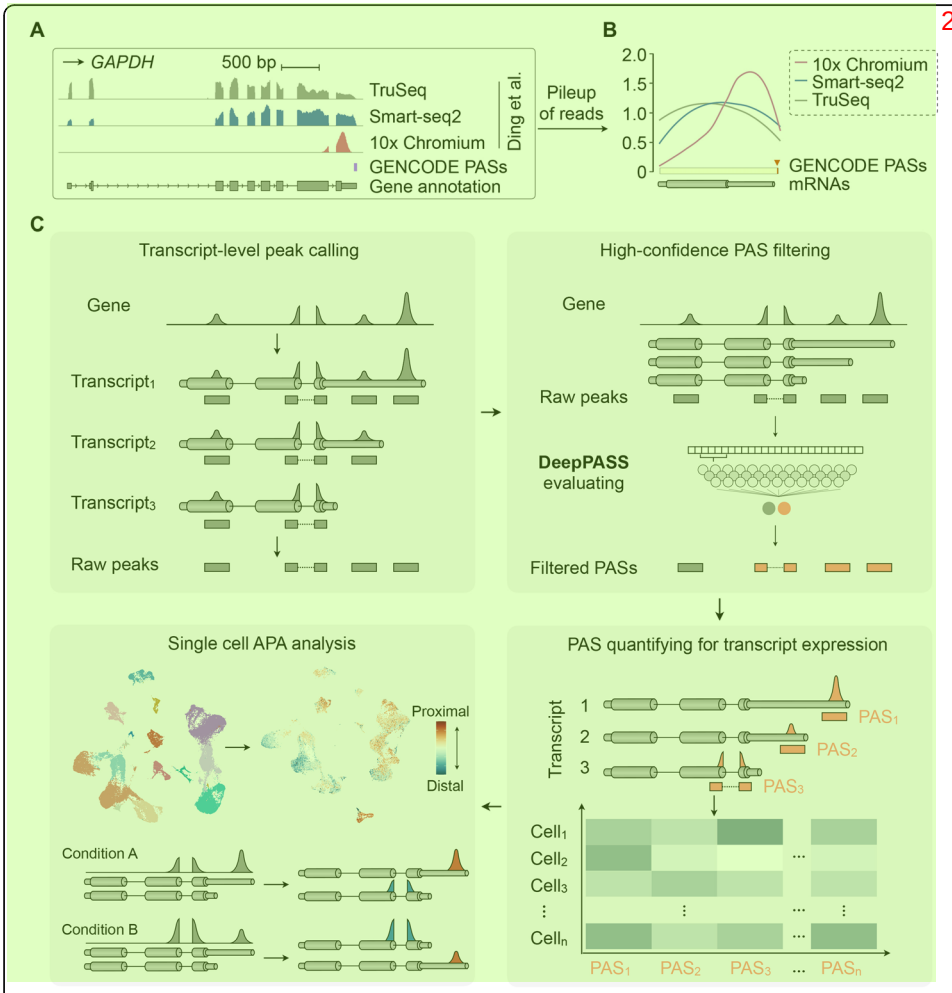

Fig. 1 Developing SCAPTURE to identify cleavage and polyadenylation sites (PASs) from 3′ tag-based scRNA-seq. **a** Comparison of human PBMC transcriptome profiling with different deep sequencing datasets. Wiggle tracks show an enrichment of 10x Chromium reads (rose) at the 3′ end of the *GAPDH* gene locus, close to its known PAS (GENCODE), while reads of TruSeq RNA-seq (gray) and Smart-seq2 (dark blue) cover the whole gene body. Data were retrieved from published PBMC TruSeq RNA-seq, Smart-seq2, and 10x Chromium [17]. **b** Distribution of deep sequencing reads on mRNA genes. Pileup of deep sequencing reads from the same published datasets (**a**) indicates enrichment of 10x Chromium reads (rose) at 3′ ends of genes, compared to coverage of gene bodies by TruSeq RNA-seq (gray) and Smart-seq2 (dark blue) data. The distribution of PASs on mRNA genes were annotated in GENCODE. **c** Schematic of a stepwise SCAPTURE pipeline for single-cell PAS calling, filtering, transcript calculating, and APA analyzing. Top left, calling peaks from 3′ tag-based scRNA-seq (the first step). Top right, identifying high-confidence PASs with an embedded deep learning neural network DeepPASS (the second step). Bottom right, quantifying PASs to represent transcript expression at a single-cell resolution (the third step). Bottom left, applying SCAPTURE to APA analysis at single-cell level (the fourth step). See "Methods" section for details

Results

Development of SCAPTURE for PAS analysis with 3′ tag-based scRNA-seq data

To utilize 3′ tag-based scRNA-seq data for PAS analysis, we developed a stepwise computational pipeline named *sc*RNA-seq *a*nalysis for *P*AS-based *t*ranscript expression *u*sed to *re*fine cell identities (SCAPTURE, Fig. 1c; "Methods" section). Briefly, SCAPTURE takes aligned bam files as input to call peaks that are close to PASs of genes (Fig. 1c, top left). These called peaks are then evaluated by an embedded deep learning method to select high-confidence PASs (Fig. 1c, top right). Next, selected PASs are quantified by UMI-tools [18] to indicate expression of transcripts according to their distinct PAS usage (Fig. 1c, bottom right). Finally, SCAPTURE identifies altered PAS usage and transcript expression among different cell types/conditions at the single-cell level (Fig. 1c, bottom left). Two features of SCAPTURE are of note. First, a deep learning neural network, named DeepPASS, is embedded in the SCAPTRUE pipeline. It is trained by selecting sequences through shifting around known PASs with stringent filtering (called stringent PASs for simplicity, "Methods" section), resulting in identification of high-confidence PASs (Fig. 2a; Additional file 1: Fig. S2A; Additional file 2: Table S1; "Methods" section). Briefly, a predicted probability ranging between 0 and 1 is obtained by the DeepPASS model for any given PAS candidate, followed by classification into a positive site group (predicted probability > 0.5) or a negative site group (predicted probability ≤ 0.5) (Fig. 2b, "Methods" section). Compared with commonly used methods that are based on fixed sequences for feature extraction (termed DeepPASS-fixed here), DeepPASS achieved a higher area under curve (AUC) value with the training set (randomly selected from 90% of stringent PASs, "Methods" section) (Fig. 2c). As a result, by combining a convolutional neural network (CNN) and a recurrent neural network (RNN) for data training, DeepPASS achieved an AUC over 0.99 when using the validation set (the remaining, independent 10% of stringent PASs, "Methods" section), substantially higher than previously reported methods, such as DeepPASTA [19] and APARENT [20] (Fig. 2d; "Methods" section). Notably, this sequence selection through shifting strategy in DeepPASS can not only generate a larger training set than using fixed sequences, boosting its accuracy in PAS, but also makes predictions less sensitive to positions, increasing its sensitivity (Fig. 2e). Profiling active motifs of 128 kernels from first convolutional layer of DeepPASS showed that it was able to capture key motifs of PASs. Top captured motifs included canonical poly(A) signal AAUAAA (Entropy: 4.76) and its variants located 25 bp upstream of the cleavage site, as well as downstream U-rich motifs and the typical CA nucleotides located at the cleavage site (Additional file 1: Fig. S2B), further suggesting that DeepPASS is able to identify high-confidence PASs.

Another key feature of SCAPTURE is that the quantitative information of identified PASs is used to represent expression of transcript isoforms with distinct PASs (Fig. 1c, bottom right). It is well known that, through the alternative polyadenylation (APA) mechanism, multiple transcripts with distinct PASs (called APA transcripts) can be produced from a single-gene locus [21–23], increasing the transcriptomic complexity of the genome. Differential expression of APA transcripts has been widely examined across different cell types, but rarely at the single-cell resolution. Since different APA transcripts harbor distinct PASs, quantified PAS values can be used to represent differential transcript expression (DTE) at given gene loci. Hence, the DTE information generated by SCAPTURE can help refine cell identity analysis beyond gene expression (Fig.

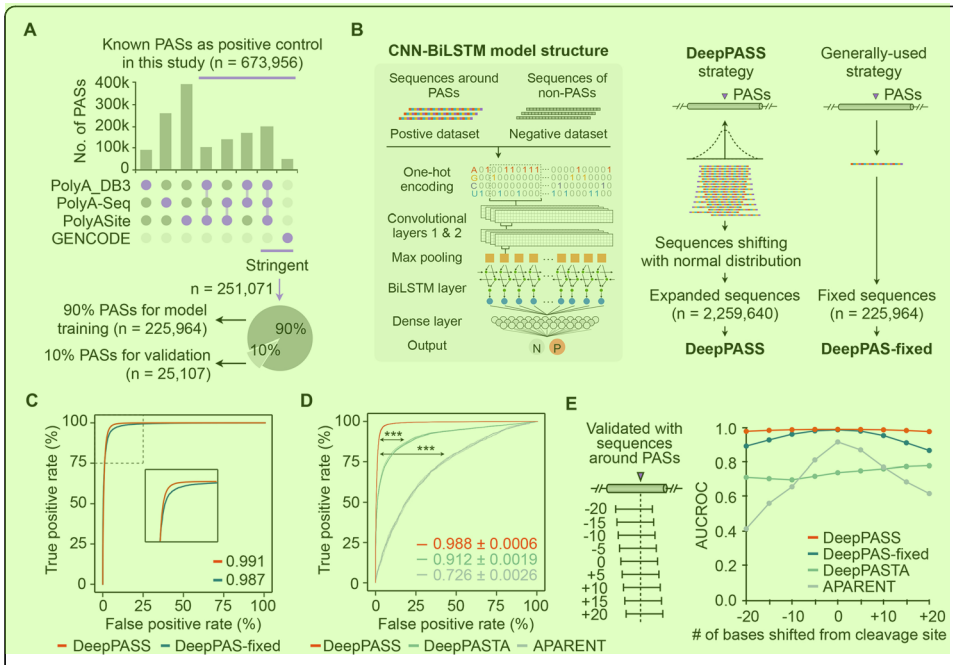

Fig. 2 Constructing the embedded DeepPASS model for position-insensitive prediction of PASs. **a** Construction of known PAS set and stringent PAS set. Previously reported PASs in at least two databases of PolyA_DB3, PolyA-seq, and PolyASite (v2.0) or in the manually examined GENCODE annotation were combined to achieve known PAS set. Stringent PAS set was further constructed from known PASs that are annotated in all three databases or in the GENCODE annotation and was split with a 9:1 ratio between a model training set and an independent validation set for DeepPASS and DeepPAS-fixed models. **b** Schematic of DeepPASS construction and evaluation. Left, data processing strategy and model architecture. Middle, a sequence shifting strategy around stringent PASs was applied to construct positive training set for establishing DeepPASS model. Right, the generally used strategy with fixed sequences around stringent PASs for DeepPAS-fixed model. See "Methods" section for details. **c** The ROC curves of DeepPASS and DeepPAS-fixed to indicate their training performance. AUC values of DeepPASS (red) and DeepPAS-fixed (blue) were shown in plot. **d** The ROC curves of DeepPASS, DeepPASTA, and APARENT on the validation set. AUC values of DeepPASS (red), DeepPASTA (green), and APARENT (purple) were shown in plot from five independent validation repeats with very low standard errors. ***$p < 0.001$, statistical significance was assessed by Student's *t* test. See "Methods" section for details. **e** Position-insensitive prediction of PASs with DeepPASS model. To assess positional tolerance of different models, 200-bp sequences shifting around the PASs in validation set with 5 bp stride were used to test accuracy of each model. The percentage represents true positive rate of each condition. DeepPASS (red) is more tolerant than DeepPAS-fixed (blue) and previously reported DeepPASTA (green) and APARENT (purple) models

1c, bottom right; "Methods" section). By contrast, typical scRNA-seq tools perform cell clustering by using the DGE information only [3].

SCAPTURE for exonic PAS analysis with human PBMC scRNA-seq data

We applied SCAPTURE to profile PASs from publicly available scRNA-seq datasets of human PBMCs based on the 10x Genomics platform (https://support.10xgenomics.com/single-cell-gene-expression/datasets) (six datasets in total; Additional file 1: Fig. S3A, B). Of the 83,390 raw peaks called in exons, 35,378 high-confidence PASs (selected by DeepPASS with predicted probability > 0.5) were identified by SCAPTURE (Fig. 3a, left; Additional file 1: Fig. S4A; Additional file 3: Table S2). Among them, 29,664 (83.8%) peaks overlapped with known PASs, which were named overlapped exonic PASs. The rest were named non-overlapped exonic PASs (Fig. 3a, right; Additional